Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, SQL, C/C++
Software Engineering & Backend: FastAPI, Flask, REST APIs, Node.js, Git/GitHub, Object-Oriented Programming
Databases & Integration: SQL, SQLite, MySQL, MongoDB, Redis, REST API Integration
Data & Business Analytics: Pandas, NumPy, Data Analysis, Exploratory Data Analysis, Statistics, Power BI, Tableau, DAX, Business Intelligence
Machine Learning: Scikit-learn, Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, Anomaly Detection
Process & Solution Design: Business Process Analysis, Workflow Design, Technical Solution Design, Design Thinking, Functional Prototyping, End-to-End Process Modeling
Engineering Practices: Data Structures & Algorithms, Data Preprocessing, PyTest, Test-Driven Development, Reproducible Experimentation

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered a Python analytics pipeline processing 50K+ transactions, combining statistical analysis, duplicate detection, vendor-velocity checks, HHI analysis, and Isolation Forest for automated anomaly detection.
- Designed a SQLite star-schema warehouse and composite 0–100 risk scoring pipeline, identifying 357 high/critical-risk transactions with automated analysis completed in under 7 seconds.
- Built a 5-page Power BI audit dashboard with 20+ DAX measures and automated pipeline tests to validate data quality, anomaly detection, and reporting reliability.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack Python application using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP to automate receipt processing and reduce processing time by 40%.
- Engineered and validated REST APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export for reliable application workflows.
- Deployed the application on Render and Vercel, implementing end-to-end workflows across AI processing, backend services, database operations, and frontend validation.

***API Rate Limiter Library* |LINK**

- Built a Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for API automation and reliable traffic control.
- Designed a modular, thread-safe architecture with pluggable storage backends to support concurrent request handling and reliable API behavior across application environments.
- Developed 30+ PyTest automated tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for reusable software development.

Customer Lifetime Value Engine |LINK

- Built an end-to-end ETL pipeline processing 1M+ customer transactions, performing data cleaning, validation, and feature engineering for predictive analytics.
- Developed BG/NBD and Gamma-Gamma models for customer lifetime value prediction, achieving an R^2 score of 0.87 on historical purchase data.
- Created interactive Power BI dashboards for customer segmentation and revenue forecasting, reducing reporting time by 80%.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** 2025